

Introduction to Java Programming

Question Bank for Assignments & Examination



Assignment Set 1 — Foundations of Java

Topics: Introduction to Java · Data Types & Variables · Operators · Control Flow · Arrays

Introduction, Data Types & Variables

- Q1. What is the difference between JDK, JRE, and JVM? Explain their roles in the Java execution process.
- Q2. What is Bytecode in Java? Why is Java called a platform-independent language?
- Q3. Explain all 8 primitive data types in Java with their size, range, and default values.
- Q4. What is the difference between implicit type conversion (widening) and explicit type casting (narrowing) in Java? Give examples.

Operators & Control Flow

- Q5. Differentiate between the logical operators && (AND), || (OR), and ! (NOT) with the help of truth tables.
- Q6. Explain the if-else ladder with a suitable real-world example (e.g., grading system).
- Q7. What is the difference between a while loop and a do-while loop? When should each be used?
- Q8. Explain the use of break and continue statements in loops with examples.

Arrays

- Q9. How do you declare, initialize, and traverse a 1D array in Java? Explain with an example of finding the largest element.
- Q10. What is a 2D array in Java? How is memory allocated for a 2D array? Explain with an example of representing a 3×3 matrix.



Assignment Set 2 — OOP, Inheritance & Advanced Java

Topics: OOP Concepts · Classes & Objects · Inheritance · Interfaces · Exception Handling · Collections · Java 8+ Features

OOP Concepts, Classes & Constructors

- Q1. Define the four pillars of OOP — Abstraction, Encapsulation, Inheritance, and Polymorphism — and give a brief real-world analogy for each.
- Q2. What is the difference between a default constructor and a parameterized constructor? What is constructor overloading?
- Q3. Explain the significance of the `this` keyword in Java. How is it used to avoid variable shadowing?
- Q4. What are static variables, static methods, and static blocks? How do they differ from instance members?

Inheritance & Interfaces

- Q5. What are the different types of inheritance in Java? Why does Java not support multiple inheritance through classes, and how is it achieved using interfaces?
- Q6. Explain method overriding and dynamic method dispatch (runtime polymorphism) in Java. What role does the `super` keyword play?
- Q7. What is the difference between an abstract class and an interface in Java? Under what scenarios would you choose one over the other?

Exception Handling, Collections & Modern Java

- Q8. Explain the exception hierarchy in Java. What is the difference between checked and unchecked exceptions? Give two examples of each.
- Q9. What is the Java Collections Framework? Compare `ArrayList` and `HashSet` in terms of order, duplicates, and performance.
- Q10. What are Lambda Expressions in Java 8? Explain their syntax and how the Stream API (`filter`, `map`, `forEach`) simplifies collection processing.